COMPARATIVE ANALYSIS OF PUBLIC TRANSPORT DATA (2019 & 2022)

1.0 INTRODUCTION

Two datasets are used for the analysis, 2019data0.csv and 2022data0.csv. The 2019 dataset refers to *daily aggregated* public transportation information by bus, tram and metro with peak and off-peak ridership and revenue information entered per day (each row representing one day). The 2022 dataset refers to *individual journey-level* data with fields such as date and time, mode, distance, duration, price (each row reflecting one trip). The two datasets combine to form a complete picture of 2019 as system-wide daily aggregates from each category and 2022 as a more detailed examination of the passenger experience.

2.0 EMPIRICAL RESULTS

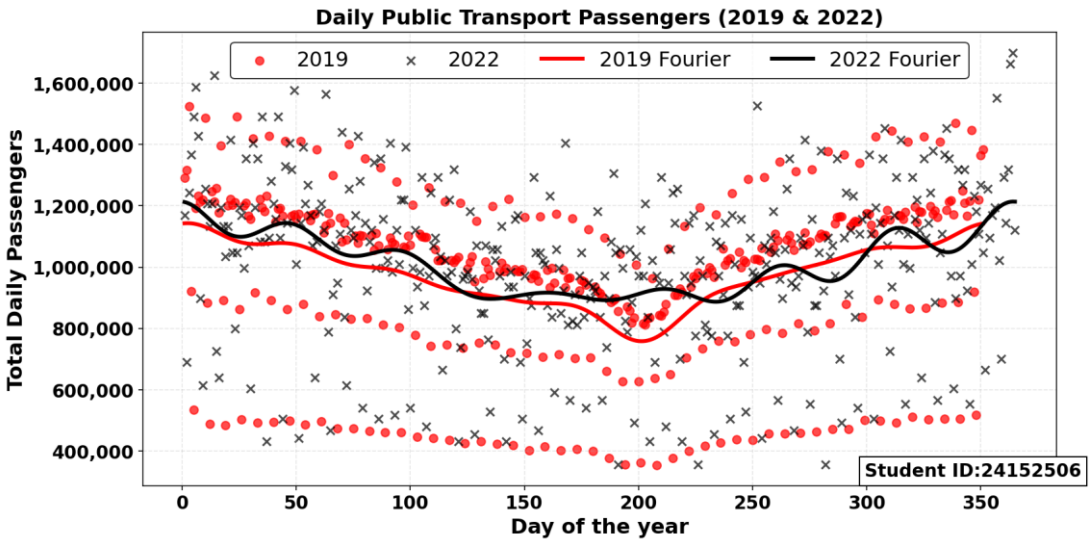

Figure 1: Daily Public Transport Passengers (2019 & 2022)

The scatter plot of daily total passengers for 2019 & 2022 is plotted against days of the year, with an 8 term Fourier series overlaid to identify any trends in the data. The dataset of 2022 is a statistical sample of passenger movement. As such, the total number of passengers per day was estimated by multiplying the number of samples by the total number of passengers on an annual basis. This guarantees that 2019 and 2022 are similar population amounts with the appropriate physical units (total daily passengers).

Observations:

- Daily passenger counts of 2019 and 2022 have more or less the same seasonal trends throughout the year.
- 2022 data is less smooth in its Fourier curve, with more day to day variability.
- The low point would be in the mid year period in both series and the upward trend would happen towards the end of the year.

Interpretation:

Fourier smoothers reveal that both 2019 and 2022 have a definite seasonal cycle with underlying trends but 2022 is more irregular. This can be an indicator of a shift in the long term seasonal organization via changes in travel behavior and schedule patterns in a post-pandemic period, not necessarily a long-term change.

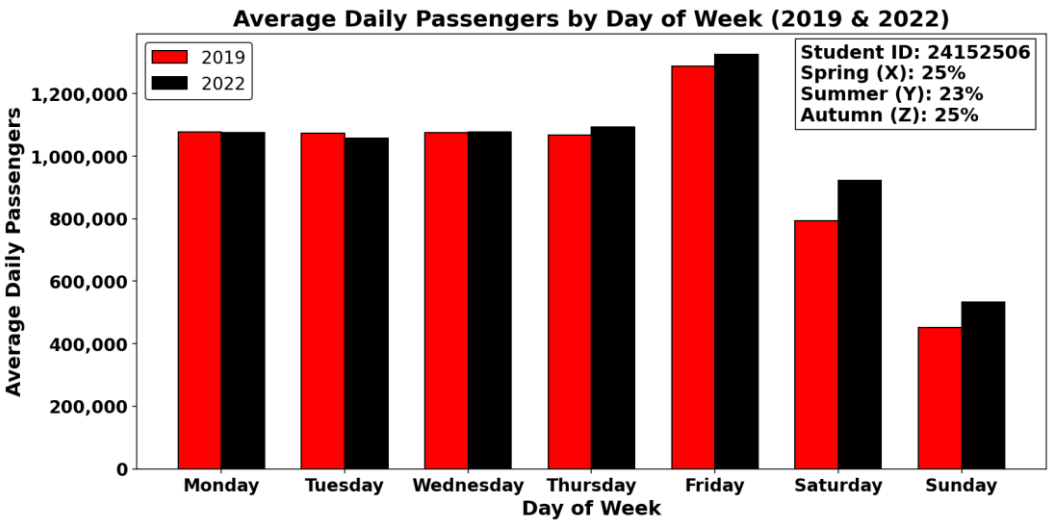

Figure 2: Average Daily Passengers by Day of Week (2019 & 2022)

Figure 2 compares the average number of passengers on each weekday in 2019 and 2022. Absolute passenger totals are used, which allows a direct comparison of weekly travel patterns across both years.

Observations:

- Passengers in 2019 show a well defined weekday pattern, more heavily used from Monday to Friday and at a lower level during Saturday and lowest on Sunday.
- 2022 weekly pattern is also far less steep and it has a diminishing distance from weekdays to weekends.

Interpretation:

The figure shows how travelling behaviour transformed from 2019 to early 2022. In 2022, the weekly pattern is noticeably flatter, with little differences in weekdays and notably higher saturday and sunday usage, suggesting more flexible travel behaviour and reduced dependence on fixed workdays.

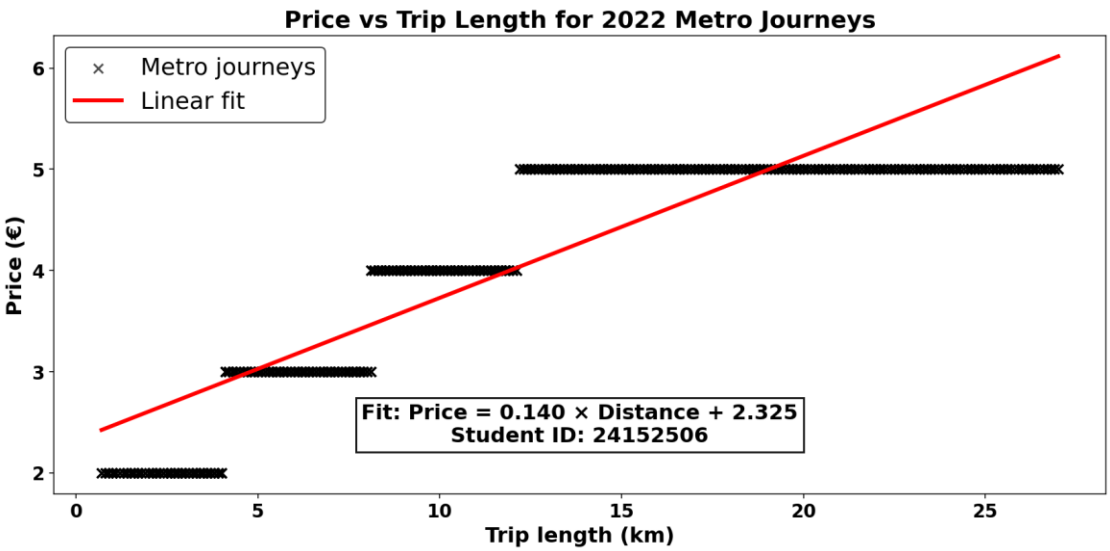

Figure 3: Price vs Distance for 2022 Metro Journeys

The scatter plot shows how trip distance (km) affects metro prices (€) during 2022 through a linear regression line that represents the data pattern.

Observations:

- As predicted, there is a clear positive linear trend: the longer your trip, the more it costs in general.
- Results have a small dispersion around the fitted line, revealing that the pricing pattern is fairly uniform.
- Shorter trips have the least dispersion between low and high price, as longer ones demonstrate more range, but still proving relatively consistent overall.

Interpretation:

The linear regression line shows that the fare system operates based on distance because price levels directly relate to the distance travelled. The points in the data set follow the line closely, which indicates that the fare system operates with consistent pricing rules that follow the expected model.

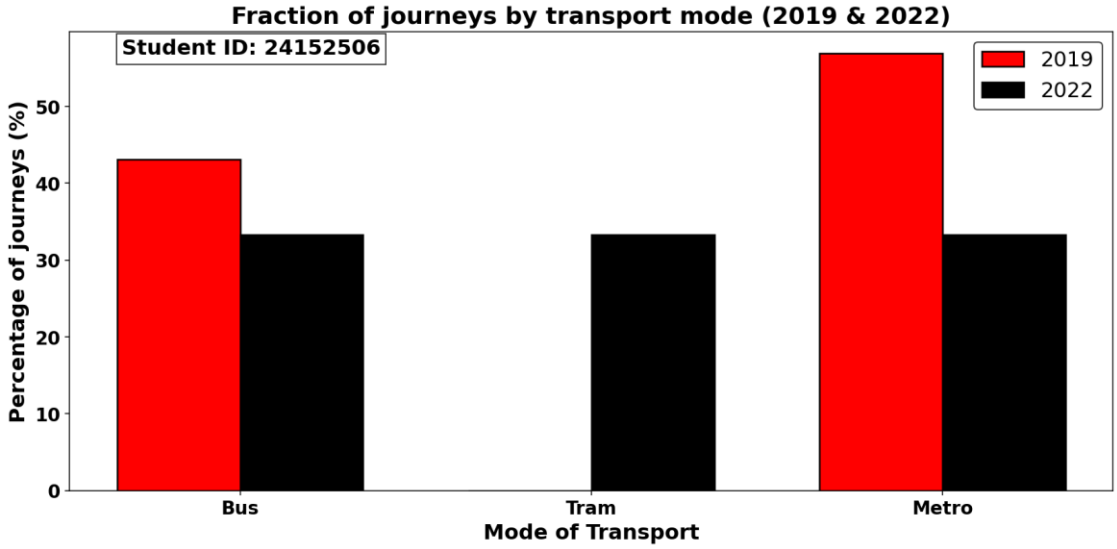

Figure 4: Fraction of Journeys by Transport Mode (2019 vs 2022)

The bar plot shows how transport modes were distributed across 2019 and 2022 through percentage values. The data shows the percentage of total recorded journeys for each year.

Observations:

- In 2019, metro is the dominant modal share and tram percentage is zero, because tram passengers are not counted in the statistics for that year.
- In 2022, bus, tram and metro are reported in more similar percentages.
- By 2022 metro usage is less frequent compared with 2019, the largest relative decline of all modes, while bus and tram use is relatively more common by 2022.

The seasonal percentages X = 25%, Y = 23%, Z = 25% represent the distribution of journeys in 2022 during spring, summer and autumn respectively

Interpretation:

Modal differences stem from data construction methods rather than changes in human behaviour because particular transportation options lack representation in specific time periods. The data shows seasonal fluctuations which remain moderate except during spring and fall months. The chart demonstrates essential points: The data contains missing information.

3.0 MATHEMATICAL METHODS

1. **Sample Expansion (2022 Estimation)**

The 2022 dataset is made up of individual journey records (sample), so the daily totals were reconstructed using the expansion factor:

$$\text{Expansion factor} = \frac{\text{Total annual passengers}}{\text{Total sample size}}$$

Estimated daily totals:

$$\text{Daily total}_{2022}(t) = \text{Sample count}(t) \times \text{Expansion factor}$$

This ensured both years are expressed in the same physical units: total daily passengers.

2. **Fourier Series**

Daily totals $\hat{y}(t)$ were smoothed using the first 8 terms of the Fourier series:

$$\hat{y}(t) = a_0 + \sum_{k=1}^{8} \left(a_k \cos\left(\frac{2\pi kt}{N}\right) + b_k \sin\left(\frac{2\pi kt}{N}\right) \right)$$

In implementation:

$$\text{FFT}[k > 8] = 0$$

3. **Linear Regression**

The fare distance relationship uses the model:

$$y = ax + b$$

Least-squares estimates:

$$a = \frac{\text{Cov}(x,y)}{\text{Var}(x)},\ b = \overline{y} - a\overline{x}$$

Where:
x = trip distance (km)
y = price (€)

4. **Seasonal Percentages (X, Y, Z)**

Let N_s, N_m, N_a be the number of journeys in spring, summer, and autumn, and N the total number of journeys:

$$X = \frac{N_s}{N} \times 100, \qquad Y = \frac{N_m}{N} \times 100, \qquad Z = \frac{N_a}{N} \times 100$$

These were computed using the journey timestamps from 2022.

5. **Mode Fractions**

For a mode math (i.e., bus, tram, metro):

$$\text{Fraction}_i = \frac{N_i}{\sum_j N_j} \times 100$$

Where math is the number of journeys (2022) or total passengers (2019).

4.0 DISCUSSION

The comparison shows that there is a significant distinction between the use of public transport in 2019 and 2022 on annual, monthly, weekly, and daily basis.

Yearly differences:

Having estimated the number of passengers in 2022 based on the representative sample totaling daily passengers, the size of the two years compares quite widely with the total annual passengers varying by approximately 3%. Thus the primary difference is not a significant change in general utilisation, but changes in daily variability and seasonal pattern, which potentially embody a change in travel behaviour after the pandemic.

Seasonal and monthly variability:

The X, Y and Z values show that autumn makes the largest percentage of the yearly journeys followed by spring and summer. This is indicative of a low seasonal trend in the use of public transport. The Fourier smoothing indicates a more pronounced seasonal cycle in 2019 than the one in 2022, which may be viewed as more erratic and may consider it a direct consequence of post-pandemic travel behaviour.

Weekly variation:

The values on Fridays are significantly higher than in other days of the week in 2019, which could be attributable to the commuter activity before the weekend. By 2022, the amount of differences between weekdays is significantly reduced, which implies lower-intensity commuting patterns. This can be affected by the heightened hybrid employment and altered travel patterns after the pandemic.

Daily variation:

In 2019, the daily passengers numbers have a very visible weekly and seasonal pattern with the daily fluctuations being more flowing throughout the year. In comparison, 2022 shows less strong and more fluctuated trends and significantly smaller volumes, which suggest decreased post-pandemic volumes of people using the transport.

Relationship between price and distance.

Figure 3 displays a positive linear association between trip distance and ticket price of the metro journeys in 2022, which validates a distance-based fare system, which is generally stable but not broadly influenced by the broader shift in passenger behaviour.

Mode of transit:

Figure 4 indicates that bus transport constitutes the highest proportion of passenger travel in both years topped by metro and tram occupies the least proportion in total. This implies that there was no significant change in relative mode preferences in 2019 to 2022.

5.0 CONCLUSION

Figures 1 - 4 show that the total use of public transport in 2022 is significantly reduced compared to 2019, and there are also changes in the daily and weekly trends. These changes can probably be attributed to post-pandemic behavioural changes, such as remote and flexible working. The low number of passengers has not entirely concealed the structural properties of the day-of-week pattern as well as the relationship between distance and fare, indicating that the underlying transport system has not been able to change its normal operations form despite the shrinking passenger demand. In general it can be seen that passenger behaviour has down scaled instead of restructuring essentially.